

Hardik Choudhary

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EDUCATION

Carnegie Mellon University

Master of Science in Mechanical Engineering – Research

Pittsburgh, PA

May 2027

Selected Coursework: Modern Control Theory, Nonlinear Control, Deep Learning, Machine Learning and AI for Engineers

Sardar Vallabhbhai National Institute of Technology

Bachelor of Technology in Mechanical Engineering | GPA: 8.03/10.0

Surat, India

May 2024

Selected Coursework: Introduction to Robotics, Mechatronics, Finite Element Analysis, Machine Design, Dynamics

SKILLS

Programming: Python, C++, MATLAB, Simulink

Robotics and Simulation: MuJoCo, Drake, Isaac Sim, Pinocchio, ROS, Webots

Embedded Hardware: Arduino, ESP32, NVIDIA Jetson Nano, Raspberry Pi

Design and Analysis: SolidWorks (PDM/PLM), ANSYS (FEA), CFD, GD&T, Design Optimization

Manufacturing: CNC Machining, 3D Printing (FDM, SLS, SLA)

RESEARCH EXPERIENCE

Carnegie Mellon University

Graduate Student Researcher, On-Orbit Servicing (Advisors: Howie Choset, Matt Travers)

Pittsburgh, PA

Aug 2025 – Present

- Extend the INSAT planner to tumbling satellites with a space-time visibility graph, raising rendezvous success from 80% to 99% over 200 randomized trials
- Develop base and arm controllers that grasp and detumble a satellite within actuation and attitude limits in Drake, MuJoCo, and Isaac Sim
- Analyze grasps for Snaker, a snake-robot gripper that wraps a satellite nozzle, showing up to 5x a parallel-jaw gripper's misalignment tolerance
- Design a 15-cable, tendon-driven Snaker and build an early 3D-printed prototype with helical tendon routing

Indian Institute of Technology Gandhinagar

Project Assistant

Gandhinagar, India

Jan 2025 – Jun 2025

- Designed a high-dexterity prosthetic foot and built a test bed to evaluate it under repeatable loading

ACADEMIC PROJECTS

Engine-Powered Drone

Sardar Vallabhbhai National Institute of Technology

Surat, India

Jan 2024 – Jun 2024

- Designed a contact-free magnetic-gear transmission driving four propellers from one engine, with a servo linkage that sets each gear gap to control propeller speed; filed a patent application
- Calibrated propeller speed against gear gap with an encoder and Arduino (75% of the ideal 6,000 RPM at the smallest gap) and used the map for feedback control that improved flight stability

RAKSH Autonomous Agriculture Robot

Team Lead, Sardar Vallabhbhai National Institute of Technology

Surat, India

Feb 2023 – Dec 2023

- Led a 5-member team building a robot that navigates polyhouse crop rows, detects plant disease, and sprays targeted pesticide
- Developed navigation using sensor fusion and vision-based row detection (ROS, Jetson Nano, ESP32) and designed a 10-chemical delivery system with servo-driven valves
- Qualified the Proof of Concept stage of Robofest-Gujarat 3.0 (Rs 1,00,000 prize) and competed in the prototype stage

Formula Student Electric Car, Team Phoenix Racing

Chassis Head, Sardar Vallabhbhai National Institute of Technology

Surat, India

Nov 2021 – Jan 2023

- Led a 30-member team through chassis design and fabrication; cut frame mass with ANSYS FEA while meeting stiffness targets
- Calculated suspension hardpoints and developed the aero package; won Concept Goals Management at Formula Bharat 2023

TEACHING AND INDUSTRY EXPERIENCE

Course Assistant, Carnegie Mellon University: AI Agents for Engineers; Applied Finite Element Analysis

Pittsburgh, PA

Intern, Banaras Locomotive Works: studied WAP-6 locomotive assembly and quality control

Varanasi, India | Jun 2023 – Jul 2023